

Two tumor suppressive loci on chromosome 10 involved in human glioblastomas.

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A number of cytogenetic and molecular analyses have revealed very frequent and extensive losses of regions of chromosome 10 in human glioblastomas. Our recent studies have demonstrated that the transfer of a chromosome 10 into human glioblastoma cells resulted in suppression of their transformed and tumorigenic phenotype. To localize the suppressive region further, we isolated and characterized certain hybrid cells that had undergone chromosomal rearrangements to yield hybrid cells retaining only various regions of the inserted chromosome 10. One series of subclones showed the loss of the majority of the long arm of chromosome 10 (10q21-10qter) and regained the ability to grow under anchorage-independent conditions, but the cells still failed to exhibit significant tumorigenicity in nude mice. Another set of subclones exhibited major deletions of large segments of the long arm of chromosome 10 (10q21-q23; 10q26-qter), yet retained certain distal alleles associated with 10q24 to 10q26. These subclones were identical in their biological characteristics to the hybrids containing an intact chromosome 10, exhibiting no growth in soft agarose or in nude mice. These results implicate the presence of two independent phenotypically suppressive regions on chromosome 10 (10pter-q11 and 10q24-q26) that are involved in glioma progression.

Post-translational modifications of PTEN and their potential therapeutic implications.

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PTEN is a tumor suppressor gene localized to human chromosome 10q23.31, a genomic region frequently lost in glioblastoma and prostate cancer. The fact that PTEN encodes a lipid phosphatase with specificity towards phosphatidylinositol-3,4,5-triphosphate renders it a gate-keeper of the phosphatidylinositol 3-kinase pathway. Numerous physiological processes have been ascribed to this evolutionarily conserved molecule including proliferation, cell size determination, survival, differentiation, and cell fate specification. Indeed, mutation in PTEN gene is the genetic cause of Cowden Syndrome. Structurally, the 54-kilodalton protein is composed of two major functional domains crucial for catalytic and membrane binding functions. Additional regulatory regions in both amino- and carboxyl-termini further dictate its structural integrity, catalytic activity, and subcellular localization. Extensive characterization of PTEN primary coding sequence has revealed a multitude of post-translational modifications that fine-tune its biochemical properties. These include phosphorylation, ubiquitination, redox modifications, and acetylation. This article aims to provide an in-depth review of the diverse post-translational modifications of PTEN, focusing on their biological relevance in both normal and cancer cells. The potential applications to cancer therapy by modulating the post-translational modifications of PTEN will also be discussed.

The importance of genomic copy number changes in the prognosis of glioblastoma multiforme.

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Glial tumors are the most common tumors of the nervous system, affecting individuals at any age. Since understanding of the molecular pathologies underlying human gliomas is still very poor, the treatment and therefore prognosis of this malignancy could not yet be improved. In order to determine whether

different glioblastoma-associated genomic aberrations may serve as prognostic markers in combination with histopathological findings, 20 primary glioblastoma multiforme tumors were screened by comparative genomic hybridization, and the results were compared with histopathological and clinical features. All tumors showed genomic copy aberrations detected by comparative genomic hybridization. Regional and numerical increases in chromosome 7 copy number were the most frequently seen abnormality (10/20 tumors), followed by loss of chromosome 10 (8/20). Both of these aberrations were associated with shorter surveillance time. Chromosome 12q amplification was detected in seven tumors. Loss of 17p, 1p, and 19q in combination was seen in three cases. One of them was a giant cell GBM, whereas the remaining two cases were still alive. Combination of chromosome 1p and 19q deletions was also seen in a case with long surveillance. According to the preliminary findings of this study, in addition to the EGFR gene, amplification of other genes on chromosome 7 and the deletion of PTEN gene and other cancer-related genes on chromosome 10 appeared important to the development of glioblastoma multiforme and were associated with poor prognosis, whereas the combination of chromosome 1p and 19q deletions seems to be an informative molecular marker for better prognosis. The clinical features and genetic alterations of primary and secondary glioblastoma multiforme should be compared in large series to clarify the effective prognostic markers; and further molecular analyses focused on chromosomes 7 and 10 will be very helpful for understanding the molecular mechanisms underlying the progression of glioblastoma.

Emerging role of PTEN loss in evasion of the immune response to tumours.

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Mutations in PTEN activate the phosphoinositide 3-kinase (PI3K) signalling network, leading to many of the characteristic phenotypic changes of cancer. However, the primary effects of this gene on oncogenesis through control of the PI3K-AKT-mammalian target of rapamycin (mTOR) pathway might not be the only avenue by which PTEN affects tumour progression. PTEN has been shown to regulate the antiviral interferon network and thus alter how cancer cells communicate with and are targeted by immune cells. An active, T cell-infiltrated microenvironment is critical for immunotherapy success, which is also influenced by mutations in DNA damage repair pathways and the overall mutational burden of the tumour. As PTEN has a role in the maintenance of genomic integrity, it is likely that a loss of PTEN affects the immune response at two different levels and might therefore be instrumental in mediating failed responses to immunotherapy. In this review, we summarise findings that demonstrate how the loss of PTEN function elicits specific changes in the immune response in several types of cancer. We also discuss ongoing clinical trials that illustrate the potential utility of PTEN as a predictive biomarker for immune checkpoint blockade therapies.

Research progress on the role of PTEN deletion or mutation in the immune microenvironment of glioblastoma.

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Recent advances in immunotherapy represent a breakthrough in solid tumor treatment but the existing data indicate that immunotherapy is not effective in improving the survival time of patients with glioblastoma. The tumor microenvironment (TME) exerts a series of inhibitory effects on immune effector

cells, which limits the clinical application of immunotherapy. Growing evidence shows that phosphatase and tension homology deleted on chromosome ten (PTEN) plays an essential role in TME immunosuppression of glioblastoma. Emerging evidence also indicates that targeting PTEN can improve the anti-tumor immunity in TME and enhance the immunotherapy effect, highlighting the potential of PTEN as a promising therapeutic target. This review summarizes the function and specific upstream and downstream targets of PTEN-associated immune cells in glioblastoma TME, providing potential drug targets and therapeutic options for glioblastoma.

Role of Ubiquitination in PTEN Cellular Homeostasis and Its Implications in GB Drug Resistance.

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Glioblastoma (GB) is the most common and aggressive brain malignancy, characterized by heterogeneity and drug resistance. PTEN, a crucial tumor suppressor, exhibits phosphatase-dependent (PI3K-AKT-mTOR pathway)/independent (nucleus stability) activities to maintain the homeostatic regulation of numerous physiological processes. Premature and absolute loss of PTEN activity usually tends to cellular senescence. However, monoallelic loss of PTEN is frequently observed at tumor inception, and absolute loss of PTEN activity also occurs at the late stage of gliomagenesis. Consequently, aberrant PTEN homeostasis, mainly regulated at the post-translational level, renders cells susceptible to tumorigenesis and drug resistance. Ubiquitination-mediated degradation or deregulated intracellular localization of PTEN hijacks cell growth rheostat control for neoplastic remodeling. Functional inactivation of PTEN mediated by the overexpression of ubiquitin ligases (E3s) renders GB cells adaptive to PTEN loss, which confers resistance to EGFR tyrosine kinase inhibitors and immunotherapies. In this review, we discuss how glioma cells develop oncogenic addiction to the E3s-PTEN axis, promoting their growth and proliferation. Antitumor strategies involving PTEN-targeting E3 ligase inhibitors can restore the tumor-suppressive environment. E3 inhibitors collectively reactivate PTEN and may represent next-generation treatment against deadly malignancies such as GB.

Loss of tumor suppressor PTEN function increases B7-H1 expression and immuno-resistance in glioma.

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Cancer immuno-resistance and immune escape may play important roles in tumor progression and pose obstacles for immunotherapy. Expression of the immunosuppressive protein B7 homolog 1 (B7-H1), also known as programmed death ligand-1 (PD-L1), is increased in many pathological conditions, including cancer. Here we show that expression of the gene encoding B7-H1 increases post transcriptionally in human glioma after loss of phosphatase and tensin homolog (PTEN) and activation of the phosphatidylinositol-3-OH kinase (PI(3)K) pathway. Tumor specimens from individuals with glioblastoma multiforme (GBM) had levels of B7-H1 protein that correlated with PTEN loss, and tumor-specific T cells lysed human glioma targets expressing wild-type PTEN more effectively than those expressing mutant PTEN. These data identify a previously unrecognized mechanism linking loss of the tumor suppressor PTEN with immuno-resistance, mediated in part by B7-H1.

PTEN loss correlates with T cell exclusion across human cancers.

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Recent evidences had shown that loss in phosphatase and tensin homolog deleted on chromosome 10 (PTEN) was associated with immunotherapy resistance, which may be attributed to the non-T-cell-inflamed tumor microenvironment. The impact of PTEN loss on tumor microenvironment, especially regarding T cell infiltration across tumor types is not well understood. Utilizing The Cancer Genome Atlas (TCGA) and publicly available dataset of immunotherapy, we explored the correlation of PTEN expressing level or genomic loss with tumor immune microenvironment and response to immunotherapy. We further investigated the involvement of PI3K-AKT-mTOR pathway activation, which is known to be the subsequent effect of PTEN loss, in the immune microenvironment modulation. We reveal that PTEN mRNA expression is significantly positively correlated with CD4/CD8A gene expression and T cells infiltration especially T helpers cells, central memory T cell and effector memory T cells in multiples tumor types. Genomic loss of PTEN is associated with reduced CD8+ T cells, type 1 T helper cells, and increased type 2 T helper cells, immunosuppressed genes (e.g. VEGFA) expression. Furthermore, T cell exclusive phenotype is also observed in tumor with PI3K pathway activation or genomic gain in PIK3CA or PIK3CB. PTEN loss and PI3K pathway activation correlate with immunosuppressive microenvironment, especially in terms of T cell exclusion. PTEN loss predict poor therapeutic response and worse survival outcome in patients receiving immunotherapy. These data brings insight into the role of PTEN loss in T cell exclusion and immunotherapy resistance, and inspires further research on immune modulating strategy to augment immunotherapy.

Loss of PTEN Promotes Resistance to T Cell-Mediated Immunotherapy.

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T cell-mediated immunotherapies are promising cancer treatments. However, most patients still fail to respond to these therapies. The molecular determinants of immune resistance are poorly understood. We show that loss of PTEN in tumor cells in preclinical models of melanoma inhibits T cell-mediated tumor killing and decreases T-cell trafficking into tumors. In patients, PTEN loss correlates with decreased T-cell infiltration at tumor sites, reduced likelihood of successful T-cell expansion from resected tumors, and inferior outcomes with PD-1 inhibitor therapy. PTEN loss in tumor cells increased the expression of immunosuppressive cytokines, resulting in decreased T-cell infiltration in tumors, and inhibited autophagy, which decreased T cell-mediated cell death. Treatment with a selective PI3K β inhibitor improved the efficacy of both anti-PD-1 and anti-CTLA-4 antibodies in murine models. Together, these findings demonstrate that PTEN loss promotes immune resistance and support the rationale to explore combinations of immunotherapies and PI3K-AKT pathway inhibitors. This study adds to the growing evidence that oncogenic pathways in tumors can promote resistance to the antitumor immune response. As PTEN loss and PI3K-AKT pathway activation occur in multiple tumor types, the results support the rationale to further evaluate combinatorial strategies targeting the PI3K-AKT pathway to increase the efficacy of immunotherapy.

Translational Nanomedicine Boosts Anti-PD1 Therapy to Eradicate Orthotopic PTEN-Negative Glioblastoma.

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Glioblastoma (GBM) is resistant to immune checkpoint inhibition due to its low mutation rate, phosphatase and tensin homologue (PTEN)-deficient immunosuppressive microenvironment, and high fraction of cancer stem-like cells (CSCs). Nanomedicines fostering immunoactivating intratumoral signals could reverse GBM resistance to immune checkpoint inhibitors (ICIs) for promoting curative responses. Here, we applied pH-sensitive epirubicin-loaded micellar nanomedicines, which are under clinical evaluation, to synergize the efficacy of anti-PD1 antibodies (aPD1) against PTEN-positive and PTEN-negative orthotopic GBM, the latter with a large subpopulation of CSCs. The combination of epirubicin-loaded micelles (Epi/m) with aPD1 overcame GBM resistance to ICIs by transforming cold GBM into hot tumors with high infiltration of antitumor immune cells through the induction of immunogenic cell death (ICD), elimination of immunosuppressive myeloid-derived suppressor cells (MSDCs), and reduction of PD-L1 expression on tumor cells. Thus, Epi/m plus aPD1 eradicated both PTEN-positive and PTEN-negative orthotopic GBM and provided long-term immune memory effects. Our results indicate the high translatable potential of Epi/m plus aPD1 for the treatment of GBM.
